

Episode B6 | *International Trade*

A tariff is a tax.

This video introduces international trade, import/export taxes/tariffs, and combinations with domestic taxes.

Notes

- Maybe reframe this as ‘open the border to trade’
- The first American steel mill to use the process was constructed in 1865 in Troy, New York. In 1875, the largest-yet steel mill, Edgar Thomson Steel Works in the Pittsburgh area, was built to use the Bessemer process, financed by industrialist Andrew Carnegie.
- The tremendous iron ore deposits around Lake Superior were located far from coal deposits, and so were shipped to ports on the southern Great Lakes that were closer to the coal mines of Pennsylvania, Ohio, Indiana, and Illinois. Large integrated steel mills were built in Chicago, Detroit, Gary, Indiana, Cleveland, and Buffalo, New York, to handle the Lake Superior ore.
- As the only major steel maker not harmed during World War II, the United States iron and steel industry reached its maximum world importance during and just after World War II. In 1945, the US produced 67% of the world’s pig iron, and 72% of the steel. By comparison, 2014 percentages were 2.4% of the pig iron, and 5.3% of the steel production.
- Although US iron and steel output continued to grow overall through the 1950s and 1960s, the world steel industry grew much faster, and the US share of world production shrank. In the 1960s, the US became a major importer of steel, mostly from Japan.
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Not all markets happen within domestic borders. In fact, the original motivation for the first story we told was trade not between two farmers but between two countries. Just like Molly and Andrew can both simultaneously improve by spe-

cializing and trading, two countries, like the US and China can improve through specialization and trade. To close the loop on Part B, we're going to examine not an international spinach market, but the international market for steel. I've written these lectures in Pittsburgh PA, which has been deeply impacted by specialization and by international trade. What we're about to discuss is a highly stylized version of the history of the international steel industry. While the true history is much more complex, a simple story can say something deeper.

Setup

Pittsburgh PA is located in a place close to ... which made it a uniquely productive location for producing steel. As technology for producing steel improved, the supply curve shifted out. And as demand for steel spiked, steel production also spiked. ... *maybe show some graphs of prices and quantities.*

Show US Steel Market, with supply slowly shifting out and demand spiking. Maybe have a timeline on the horizontal axis.

During WWII, the Monogahela River basin, a few miles from here, produced something like one third of the steel in the US, to the extent Pittsburgh became a strategic target during WWII.

In the following decades, the international steel market started catching up. Eventually Chinese steel producers, among others, started undercutting the big US steel producers.

Show Chinese Steel Market, on the right, with a slow increase in the supply curve as time moves.

While there were changes in demand during this time, the main story has to do with the global supply of steel, so we'll leave the demand for steel in both countries unchanged.

If you were a US builder, given the choice between US steel or Chinese steel, you would pick the less expensive option. As Chinese steel became less expensive

than domestic US steel, even after importing across the Pacific, a US builder may choose to import steel from China instead of buying domestically.

We'll call the price of Chinese steel in the US market ***Global Price***.

Show P_g on both graphs.

If you were a US producer, seeing that US builders would not buy from you at the old prices, you would be forced to lower your prices to the global price. If you didn't no one would buy from you.

This means domestic sellers now face the lower global prices. And as we know from the law of supply, as prices go down, quantity supplied also goes down. Pittsburgh's local economy in this era was so heavily dependent on the steel industry, as international competition grew, Pittsburgh's population began not only to not grow, but the city saw a dramatic decline in population as steel jobs left the city.

Show the supply curve for China shifting out, moving the price global down, moving the US quantity supplied down

At the same time that US domestic quantity supplied began to fall with prices, US domestic quantity demanded began to rise, due to the law of demand. While lower global prices devastated Pittsburgh's industrial base, lower prices also made it easier to build things. If you're trying to build a bridge using steel, the fact that you can get steel for less means you're able to build bridges at lower cost. This also means you'll build more bridges. Many more bridges can be built because the price is lower, which is a very good thing.

We can see the total imports, which is the difference between the domestic quantity supplied and quantity demanded. People want Q_d but only Q_s is being supplied by domestic sellers, so we have to import the difference. We'll call these imports.

Add imports

On the Chinese side, at the global price, domestic buyers want Q_d but domestic sellers will sell Q_s . Chinese sellers will export the difference. We'll call these exports. You can see that in our simple story, Chinese exports are equal to US imports, which must be true.

Add exports

On the Chinese side we've also noticed something important. With
So domestically, overall, we see seller's hurt by the lower prices, but

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Lets do a little extension to bring together some of the ideas we've developed.

Lay out the recent ideas and models on the screen

Last video we left off thinking about taxes in competitive private goods markets.
We showed that the effects are identical whether we tax the buyer or the seller.
So we simplified our tax model by instead using a tax wedge.

All this took place within an environment where national borders didn't matter.
This video will be somewhat unique in this class.

We're going to consider a simple modification to our supply and demand model with taxes, in which we have a domestic market where we can import and export goods.

And we're including this model because to really understand it requires understanding both quantities and prices for both buyers and sellers.

This class isn't about international trade, but in many ways this model captures many relationships we've left unrelated.

We worked with price controls, where we set a legal interval of prices, which means quantity supply and demanded differ.

And we've set taxes, where the buyers price and the sellers' price differ.

In this model, we're going to flex both these muscles simultaneously.

So first, let's set up our domestic market of supply and demand.

This gives us a domestic price and quantity, which we'll just denote with P and Q .

But what happens if we open our domestic market up to the world? We're going to make an assumption here.

We're going to say that our domestic market is small relative to the rest of the world.

This means what our domestic buyers and sellers do doesn't impact the global prices.

Relative to domestic prices, the world price, which we'll denote P subscript w , can either be higher, lower, or equal.

If the world price is above the domestic price, instead of selling domestically, sellers can sell internationally and do better.

So no seller will want to sell at a price below the global price, which we'll label P subscript w .

Since the price is higher, the quantity supplied will also be higher.

But this change in seller behavior impacts the domestic market.

Buyers can now buy from the world market, but since the world price is above the domestic price, buyers receive no benefit from the international market.

In fact, domestic sellers are no longer willing to sell at a price below the global price.

So buyers now face the global price as well. Since the price is now higher, quantity demanded falls.

Since sellers are selling to the world market, they are able to sell more than the domestic quantity demanded.

And that's what we call exports.

And this is true at a range of prices.

Then if the global price is below the domestic price, buyers can buy off the global market instead of the domestic market, since the global price is lower.

This means sellers cannot sell at a price above the global price.

And in fact must sell at the global price on the domestic market.

And this is true at a range of prices.

So what you can see is that incentives for the buyers and sellers push the domestic price to be equal to the global price.

And the difference in quantity supplied and quantity demanded at this price is what we call the imports or exports.

This concept rhymes with our model of price controls.

Except the difference between quantity supplied and demanded is compensated for by the global market with imports and exports, instead of excess and shortage.

Welfare in this model probably works with your intuition.

Where before we constrained surplus to the region above quantity exchanged, we now have two quantities: one for sellers and one for buyers.

As before, to find CS, we look at the area below the demand curve, above the price, and inside the quantity bought.

And to find PS, we look at the area above the supply curve, below the price, and inside the quantity sold.

At this point, there's no government revenue or DWL.

Here one can see that opening the domestic market to trade increases total surplus.

In this model, let's introduce a tax on international trade: a tariff.

A tariff is typically imposed on imports: goods produced internationally and bought domestically.

But exports can be tariffed, and imports from countries importing these goods can be tariffed.

So what happens when we impose a tariff on imports?

Since buyers now face the global price, but with the tariff included, the price buyers face is now higher, and by the amount of the tariff.

This increases the sellers price as well.

Both these changes are reflected in CS and PS.

But we've also introduced government revenue.

The government is receiving a tariff on each item being imported.

So we multiply the imports by the tariff to find the government revenue.

And I promised deadweight loss is not always a triangle.

Here deadweight loss, the loss in total surplus from the maximum, is the area above the lost quantity, which happens to be lost imports, and between the demand curve and the price and supply curve and the price.

And this model applies to an export tariff as well in a similar way.

The domestic price is the global price with the tariff, which gives us producer and consumer surplus.

Government revenue is the tariff times the exports.

And deadweight loss is the lost surplus, which is the area above lost exports and between the price and the demand curve, and the price and the supply curve.